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Development of a relativistic one-dimensional model including quantum electrodynamics effects

Timothée Audinet

Dipartimento di Chimica, Biologia e Biotecnologie
University of Perugia, Italy

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Introduction

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- Includes the electron spin and spin-orbit coupling

Beyond relativistic effects

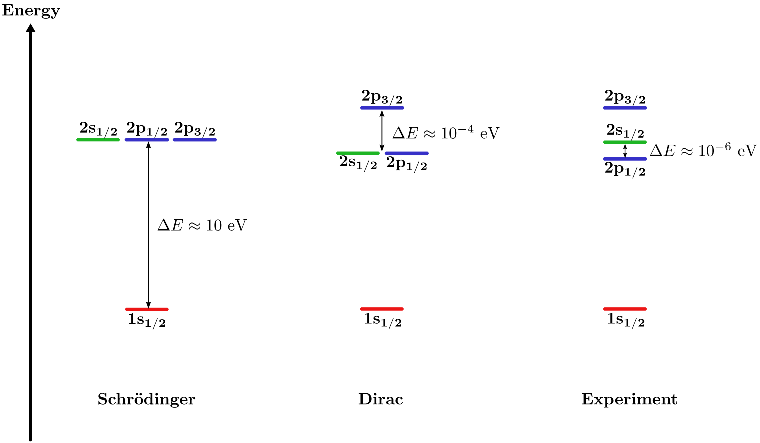

Figure 1: Lamb shift in hydrogen spectra

Non-relativistic vs Relativistic hydrogen-like spectrum

Non-relativistic spectrum

$$\boldsymbol{H}_Z = \frac{\vec{p}^2}{2m} - \frac{Z}{|\mathbf{r}|} \tag{1}$$

Spectrum:

- If $\mathscr{E} > 0$: continuum
- If $\mathscr{E} < 0$: bound states

Relativistic spectrum

$$\boldsymbol{D}_Z = c\,(\vec{\boldsymbol{\alpha}} \cdot \vec{p}) + \boldsymbol{\beta}\, mc^2 - \frac{Z}{|\mathbf{r}|}\mathbb{I}_4 \tag{2}$$

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- If $\mathscr{E} \in (-\infty, -mc^2] \cup [mc^2, +\infty)$: continuum
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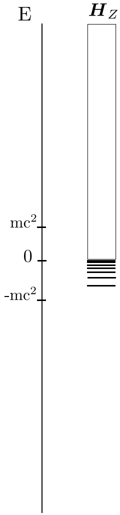
$\Rightarrow$ Negative continuum spectrum



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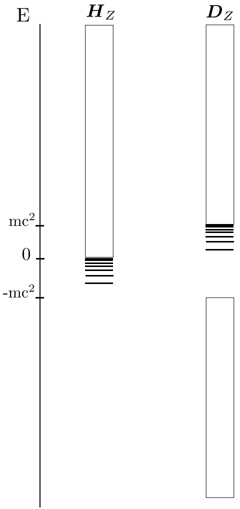

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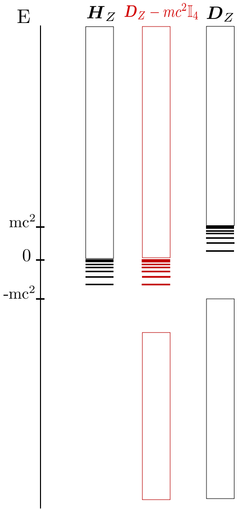
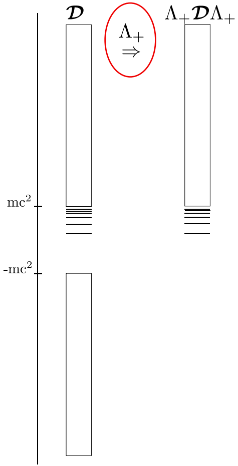
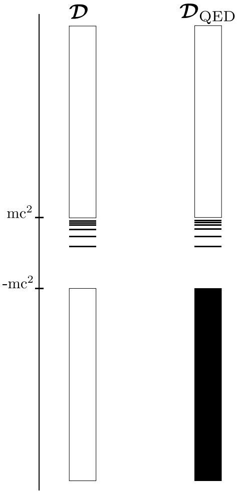


- No-pair approximation



- QED description



Motivations for the thesis

- We need accurate QED calculations for high precision spectroscopy
- So far, state of the art QED for many-electrons systems is based on no-pair approximation
- Next challenge in relativistic quantum chemistry: can we go beyond the no-pair approximation?

Goals

- Understand the divergences appearing in QED,
- Put relativistic quantum chemistry on a deeper theoretical ground,
- Develop ab initio methods based on QED.

To do list:

- [ ] Study the 1D hydrogen-like system and compute the QED effects.
- [ ] Implement QED in a finite basis set.
- [ ] Use our insight to expand this to the real 3D systems.

The 1D Operator

Free 1D Dirac operator

$$\mathbf{D}_0(x) = -\mathrm{i}c\sigma_x \frac{\mathrm{d}}{\mathrm{d}x} + \sigma_z mc^2 \tag{3}$$

Lapidus, AJP, (1983) ; T. Audinet, J. Toulouse, JCP, (2023)

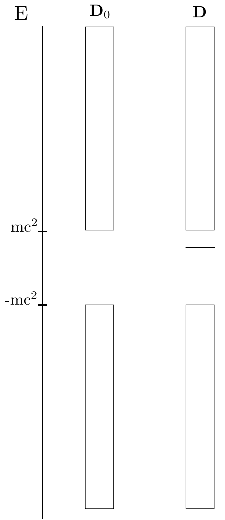
Dirac eigenvalue equations

$$\mathbf{D}_0(x)\psi_p^0(x) = \varepsilon_p \psi_p^0(x) \tag{4}$$

- Continuum: $\varepsilon_k = \pm\sqrt{m^2c^4 + k^2c^2}$

$$\left(\mathbf{D}_0(x) - Z\delta(x)\right)\psi_p^Z(x) = \varepsilon_p \psi_p^Z(x) \tag{5}$$

- Bound State: $\varepsilon_b < mc^2$
- Continuum: $\varepsilon_k = \pm\sqrt{m^2c^4 + k^2c^2}$

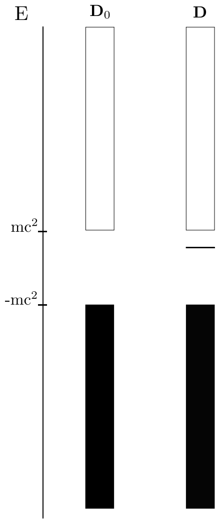


- Spontaneous creation of electron positron pairs due to the external potential
- Creates a charge density that will interact with the bound states through the two-electron interaction

Vacuum Polarization Density

$$n^{\mathrm{vp}}(x) = \sum_{\varepsilon_p<0} \psi_p^{Z\dagger}(x)\psi_p^{Z}(x) - \sum_{\varepsilon_p<0} \psi_p^{0\dagger}(x)\psi_p^{0}(x)$$

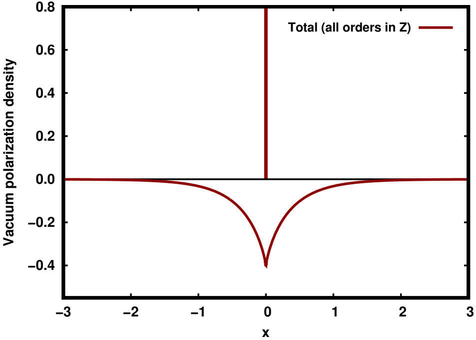
- In 3D this quantity diverges, needs to be renormalized
- We want to have a better understanding of this quantity and its influence on the energy spectrum



Vacuum Polarization Density

$$n^{\mathrm{vp}}(x) = \sum_{\varepsilon_p<0} \psi_p^{Z\dagger}(x)\psi_p^{Z}(x) - \sum_{\varepsilon_p<0} \psi_p^{0\dagger}(x)\psi_p^{0}(x) = \mathcal{N}_0^{\mathrm{vp}}\delta(x) + n_{\mathrm{reg}}^{\mathrm{vp}}(x) \quad (6)$$

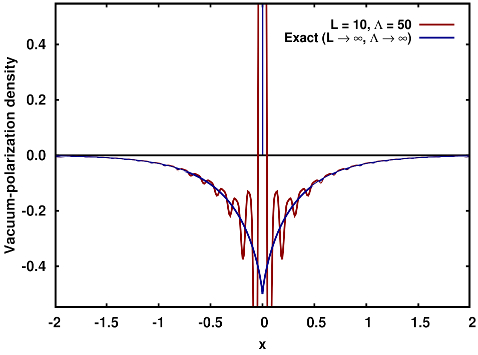
Audinet, Morellini, Levitt, Toulouse, JPA, (2025)



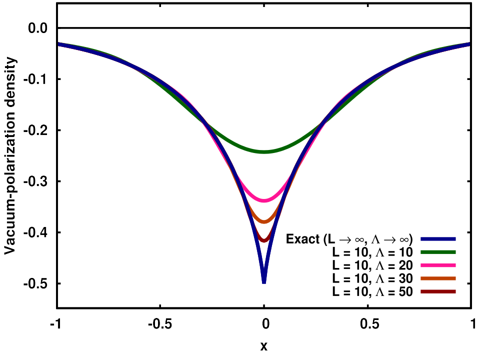
In a basis... convergence issues?

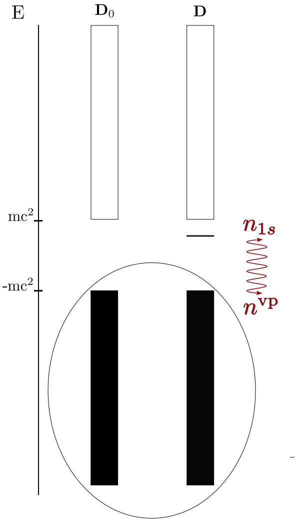
Plane-waves basis

$$\forall\, x \in \mathbb{R}, k \in \frac{2\mathbb{Z}\pi}{L}, |k| \leq \Lambda, \quad \chi_k(x) = \frac{1}{\sqrt{L}} e^{ikx} \tag{7}$$

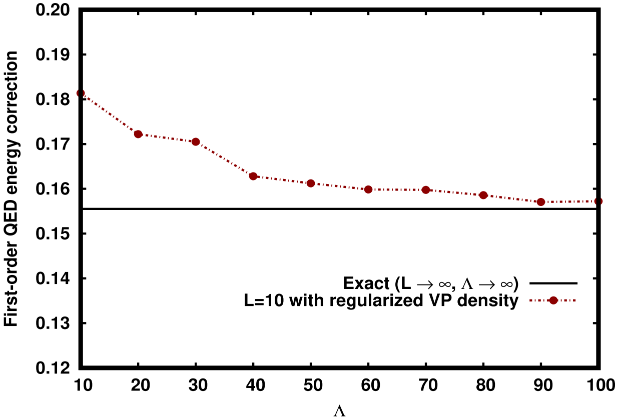


$\mathcal{FT}$
$\Longrightarrow$





δE^{vp} $\Longrightarrow$



Conclusion

To do list

- ✓ Study the 1D hydrogen-like system and compute the QED effects.
- ✓ Implement eQED in a finite basis set.
- ☐ Use our insight to expand this to the real 3D systems (in progress).

- We developed an effective QED theory including electron-positron pairs.
- Some singularities appears in a finite basis.
- This 1D model help us to understand this problem.
- Promising results for the 3D but limited by the size of the basis.

Thank you for your attention!

- Julien Toulouse
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- Enrico Ronca
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- The whole lab

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QED Spin